

Automated Circuit Discovery for NIDS

Mechanistic Interpretability for SOC-Oriented Security Research

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Presentation Outline

- 1 Motivation & Background
- 2 Technical Framework
- 3 Research Applications
- 4 SOC-Oriented Integration
- 5 Conclusion

Motivation: The Interpretability Gap in Network Security

- Modern network security systems increasingly rely on **deep learning** models for automated threat detection.
- **The Black-Box Problem:** These models generate alerts, but their **internal decision logic** is opaque to human analysts.
- **Operational Impact:** Security teams face **high false-positive rates**.
- **Key Challenge:** Understanding *why* a model raises an alert, not just *that* it raises one.
- Motivation: **causal interpretability** > correlational explanations.

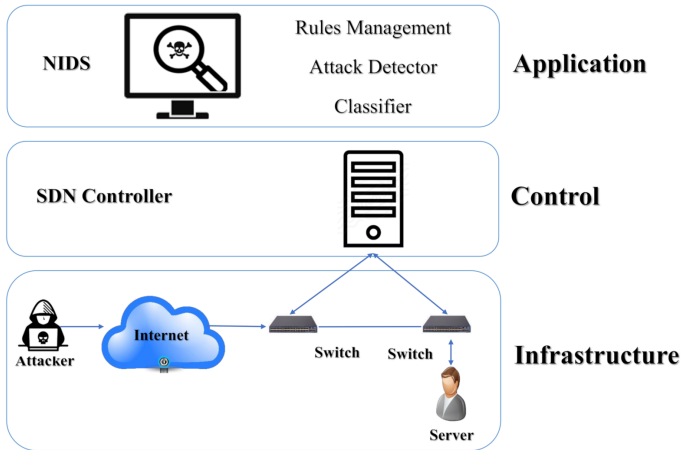
What is a Network Intrusion Detection System (NIDS)

- A **Network Intrusion Detection System (NIDS)** monitors network traffic to identify **malicious activities or intrusions**.
- It analyzes **packets, flows, and traffic patterns** traversing the network.
- NIDS raises alerts when observed traffic deviates from **normal or expected behavior**.
- It supports **Security Operations Centers (SOC)** in detecting attacks such as DoS, scanning, malware, and data exfiltration.

Key idea:

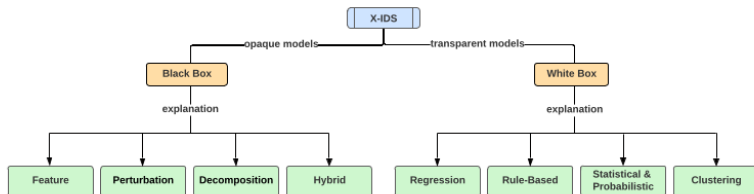
“Is this network traffic normal or malicious?”

Network Intrusion Detection System (NIDS)



High-level architecture of a Network Intrusion Detection System.

Explainability Landscape for Intrusion Detection



High-level taxonomy of explainability approaches for intrusion detection systems

Interpretability Methods in NIDS: A Comparison

Aspect	Feature Attribution (SHAP / LIME)	Rule-Based XAI	ACDC (Proposed)
Explanation Level	Input features	Human-defined rules	Internal computational components
Causality	Correlational	Assumed by rules	Causal (via activation patching)
Model Internals	Ignored	Not applicable	Explicitly analyzed
Attack-Specific Logic	Weak	Manual	Automatically extracted
SOC Usefulness	Limited	High but rigid	High and operationally actionable
Forensic Value	Low	Moderate	High (circuit-level audit)

Applying ACDC to NIDS: Core Idea

- In NIDS, the target behavior is an **attack detection decision** (e.g., SYN flood, scan, exfiltration).
- ACDC treats the trained NIDS as a **signal-processing system** that transforms raw traffic statistics into security decisions.
- The objective is to identify the **minimal internal computation** that maps:
 - traffic patterns (rates, flags, temporal bursts)
 - to an attack-specific alert.
- The extracted circuit reveals **which traffic characteristics** and **which internal interactions** the NIDS truly relies on.

Mapping ACDC Concepts to NIDS

ACDC Concept	Interpretation in NIDS
Behavior	Specific attack type (e.g., DDoS, Scan, Brute Force)
Computational Graph	NIDS model layers and feature interactions
Activation Patching	Replacing malicious traffic activations with benign baselines
Edge Pruning	Removing causally irrelevant feature pathways
Circuit	Minimal computation responsible for attack detection
Faithfulness Metric	KL divergence between full and pruned model outputs

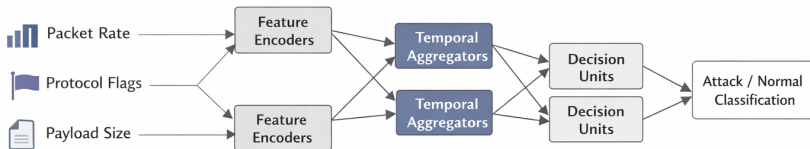
The NIDS Computational Graph

- The trained NIDS is modeled as a **directed computational graph**.
- **Nodes (V)**: internal processing units such as feature interaction layers, temporal aggregation modules, attention heads, or hidden neurons.
- **Edges (E)**: linear information pathways that propagate traffic signals (e.g., packet rates, flags, burst patterns) across the network.
- **Objective**: identify and retain only those edges that are **causally necessary** for a specific attack detection behavior.

This abstraction enables circuit discovery across different NIDS architectures (Transformer, RNN, CNN).

System Architecture: ACDC-Enabled NIDS

NIDS as a Computational Graph



Step A: Task and Dataset Setup

To find a circuit, we must define the behavior:

Task Definition

Detecting SYN Flood attacks within the UNSW-NB15 dataset.

- **Clean Dataset:** Malicious traffic samples.
- **Corrupted Dataset:** Normal baseline traffic or mean-ablated features.

Behavior is quantified using the logit difference between Attack and Normal classes.

Datasets for Attack-Specific Circuit Discovery

(a) Basic statistics of datasets

Dataset	Labels	Features	Samples
CICIDS-2017	7	78	2.78M
RoEduNet-SIMARGL2021	3	29	31.43M
NSL-KDD	5	41	148K

(b) Distribution of samples across attack types

Dataset	Normal	DoS	PortScan	Brute	Web	Bot	Probe	U2R
CICIDS-2017	84.4%	9.1%	5.7%	0.5%	0.16%	0.07%	–	–
RoEduNet-SIMARGL2021	62.2%	24.5%	13.3%	–	–	–	–	–
NSL-KDD	48.0%	35.0%	–	–	–	–	10.1%	0.47%

These datasets enable attack-specific behavior isolation and evaluation of circuit faithfulness under realistic class imbalance.

Step B: Pruning Criterion (τ)

ACDC removes an internal connection ($w \rightarrow v$) if it does not causally influence the attack detection outcome.

$$D_{KL}(G \parallel H_{\text{new}}) - D_{KL}(G \parallel H) < \tau$$

- G : Full NIDS model predicting attack vs. normal traffic.
- H : Current candidate circuit under evaluation.
- D_{KL} : Change in output probability distribution.

KL divergence ensures the circuit preserves the **decision logic**, not merely classification accuracy.

Step C: Activation Patching

- Internal activations from **corrupted (benign) traffic** are injected into the forward pass of **malicious traffic**.
- If the attack prediction remains unchanged, the patched connection is **not causally required** for detection.
- This allows ACDC to distinguish **true attack logic** from incidental correlations.
- **Example:** replacing SYN packet-rate activations with statistics from normal background traffic.

Research: Generalization vs. Spurious Correlation

- Does the train NIDS learn **attack-relevant traffic patterns** (e.g., TCP flags, packet-rate bursts, temporal structure)?
- Or does it rely on **incidental identifiers** (e.g., source IPs, ports, or dataset-specific artifacts)?
- ACDC reveals whether the extracted circuit encodes **robust attack semantics** or shortcut-based correlations.
- This analysis helps distinguish **true generalization** from spurious detection behavior.

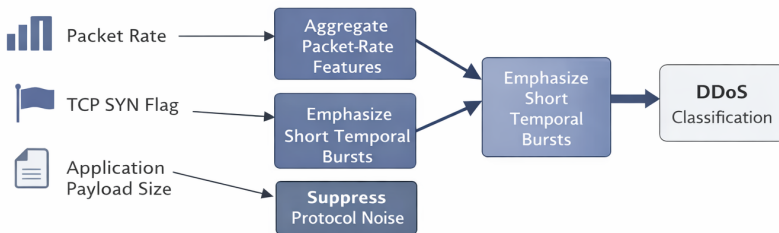
Example: Circuit Discovery for DDoS Detection

- Consider a trained NIDS tasked with detecting **volumetric DDoS attacks**.
- ACDC isolates a **minimal detection circuit** that:
 - aggregates packet-rate signals across short time windows,
 - emphasizes sudden temporal traffic bursts,
 - suppresses protocol-specific and benign background noise.
- This circuit constitutes the **causal computation** sufficient for triggering a DDoS alert.

ACDC-Based Circuit Extraction for DDoS Attacks

Example: Circuit Discovery for DDoS Detection

ACDC isolates a DDoS-specific circuit within the trained NIDS



Technical Challenge: Scalability

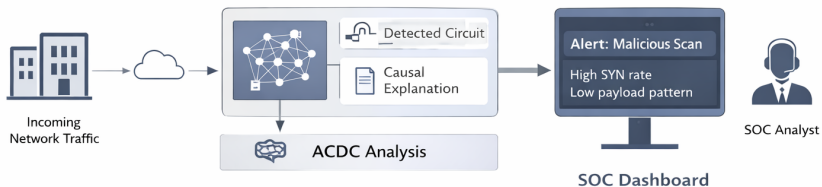
- Activation patching is computationally expensive for high-throughput network traffic.
- **Research Focus:** Attribution patching to approximate circuit importance with significantly reduced computational cost.
- As a result, ACDC is currently suited for **offline analysis and post-incident investigation**, rather than inline detection.

Failure Case: The OR-Gate Problem

- A NIDS may contain **redundant internal pathways** that independently support the same attack detection.
- If removing one pathway does not degrade detection performance, ACDC may incorrectly prune both during causal testing.
- This does not indicate model failure, but highlights **redundancy in the learned detection logic**.
- Addressing such redundancy remains an open research problem in circuit discovery methods.
- This is a limitation of greedy causal pruning, not a model error.

SOC Integration: System Design

SOC-oriented Explainable Alerts



SOC Integration: Transparent Alerts

Instead of: *Alert: Anomaly detected*

ACDC-Enhanced Alert

Logic: Observed high SYN count combined with low data payload patterns.

Causal confidence: 99.2%.

Circuits are summarized, not shown raw, to analysts.

Automated Forensics for Tier-3 Analysts

- For each detected attack, ACDC extracts the **internal detection circuit** used by the NIDS.
- Analysts can inspect the **behavioral logic** (e.g., traffic patterns and internal interactions) that triggered the alert.
- This supports **faster root-cause analysis** for advanced and previously unseen threats.

Logic-Based Filter Tuning and Evaluation

- Circuit inspection identifies causes of false positives.
- Model refinement is applied only to implicated sub-components.
- Evaluation metrics:
 - KL divergence under ablation,
 - detection accuracy preservation,
 - circuit sparsity.

Evaluating ACDC-Based NIDS: Faithfulness

- **Faithfulness:** Does the extracted circuit behave like the full NIDS?
- **KL Divergence:** Low divergence indicates preservation of detection logic.
- **Logit Margin:** Drop in attack–benign confidence reveals causally important edges.
- **Sparsity:** Fewer retained edges imply higher interpretability.

- **Recovery Quality:** Does the circuit retain true attack-relevant logic?
- **ROC-AUC:** Measures separation between relevant and irrelevant components.
- **Clean vs Corrupted Traffic:** Correct circuits detect attacks only when malicious features are present.
- **Q1 / Q2 Check:** Correct logic retained, spurious logic excluded.

Building Operational Trust

- **Auditability:** ACDC provides a causal explanation for each intrusion detection decision.
- **Accountability:** Enables post-incident review and justification in regulated and security-critical environments.
- **Inspectability:** Improves analyst confidence by making AI-driven decisions transparent rather than opaque.

- Framed network intrusion detection as a **mechanistic interpretability problem**, rather than a black-box prediction task.
- Applied Automatic Circuit Discovery (ACDC) to extract **causal, attack-specific computational circuits** from trained NIDS models.
- Demonstrated that circuit-level explanations enable **auditable and trust-oriented decision support** for SOC operations.

- Scalable and approximate ACDC methods for high-speed network environments.
- Circuit discovery under redundancy and overlapping attack logic.
- Circuit-to-rule translation for deployment in SOC workflows.
- Behavioral analysis of zero-day attacks using circuit-level explanations.
- Human-in-the-loop refinement of interpretability for trusted NIDS.

Thank You!